[NEW RESULTS]-KDD-1060

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1 Additional Experiment Results to Reviewer gq6s

2. Interpretability Validation (W1, W5)

Metric guide. For the real-data faithfulness metrics you mentioned, `I(20)` is the stricter low-budget score and `I(100)` is its full-budget counterpart. Higher values mean that the highlighted temporal evidence remains more predictive, while lower rank is better.

Table 1 summarizes the real-data faithfulness results under a fixed masking-and-retraining protocol. The central question here is not full semantic interpretation, but whether the top-ranked temporal regions continue to carry predictive signal on real data. Under the stringent `I(20)` threshold, AnchorMoE achieves **0.6419**, exceeding both the strongest TS-specific baseline (FIT, **0.5760**) and the strongest generic baseline in the summary view (IG, **0.6189**). Under `I(100)`, AnchorMoE reaches **0.8620**, and its gradient-free variant reaches **0.8667**, both outperforming Dynamask (**0.8284**). The appropriate reading is therefore stronger temporal faithfulness on real data: the highlighted evidence remains predictive and compact, although these results do not by themselves establish full domain-semantic interpretability.

Figure 1 provides a representative NATOPS case study to complement Table 1. Its role is qualitative but specific: it shows that the quantitative faithfulness gains are accompanied by a compact temporal focus rather than diffuse highlighting over the full sequence.

2 Additional Experiment Results to Reviewer Hjgq

1. Principal Explanation (W1, W3, W4, W5)

Metric guide. For the faithfulness metrics you mentioned here, `I(20)` and `I(100)` measure how much predictive signal is preserved by the highlighted evidence under a strict and a more permissive evidence budget, respectively; higher values indicate stronger temporal faithfulness, and lower rank is better.

Table 2 isolates the role of gradients in UADB. Under the real-data faithfulness protocol, the gradient-free variant of AnchorMoE still exceeds IG on mean `I(100)` (**0.8667** vs. **0.8384**), while the standard AnchorMoE model remains stronger under the more restrictive `I(20)` threshold (**0.6419** vs. **0.6189**). This pattern is more consistent with gradient magnitude acting as a training-time calibration signal than as the final explanation returned at inference. (Refer to Table 1 for complete dataset-level statistics.)

Table 2: Gradient-role control analysis. Higher is better for `I(20)` and `I(100)`; lower is better for rank. The best and second-best values in each column are highlighted.

Method	I(20)	Rank	I(100)	Rank
AnchorMoE	**0.6419**	**2.57**	**0.8620**	**2.29**
AnchorMoE-gradient-free	**0.6338**	3.14	**0.8667**	**2.43**
IG	0.6189	**2.71**	0.8384	3.14

2. Internal Mechanism (W2, W7, W8, W9)

Metric guide. For the utilization statistics you asked about, active expert rate is the fraction of experts in use, max expert share is the busiest expert's share, and effective experts is the diversity-adjusted number of experts in use. For κ, mean / low / high summarize gate sparsity; lower redundancy means less overlap among routed specialists.

Tables 3–6 separate three mechanism questions that are often conflated in MoE discussions: whether utilization is non-trivial, whether routed content becomes redundant, and whether the confidence gate collapses to a trivial regime. Table 3 reports aggregate statistics across 29 datasets. Tables 4 and 5 then show that both routing concentration and κ sparsity are strongly dataset-dependent rather than fixed to one global pattern. Finally, Table 6 shows that removing DPAR increases inter-anchor redundancy from **0.3488** to **0.7213**, which is more consistent with DPAR acting as a redundancy-reducing structural prior than as a traffic-balancing device.

Table 7 reports targeted internal-design diagnostics. Removing the sample-level relevance cue lowers mean Acc from **0.9686** to **0.7084**, whereas removing the spectral cue does not reduce the block mean. This pattern is consistent with the mechanism design in the main paper: the relevance cue injects sequence-level context into routing, while the spectral branch acts as a complementary stability-oriented cue rather than the dominant discriminative component in this diagnostic. Table 11 reports the binding-specific diagnostics: removing confidence weighting yields an all-one κ regime and lowers mean `I(20)` from **0.8953** to **0.8819**. This is consistent with the UADB design in the main paper, where routing determines where evidence is processed, while confidence-weighted binding determines how strongly routed evidence should influence the final decision. Together with Table 6, these diagnostics provide empirical support for the query design, redundancy control, and confidence-weighted binding.

Detailed sensitivity analysis (W9) is provided in Section 4.

3 Additional Experiment Results to Reviewer gMhY

1. Scalability Analysis & Runtime (W1)

Metric guide. For the scalability quantities you mentioned, throughput is measured in inference-plus-explanation iterations per second, so it already charges post-hoc explanation cost to the black-box baselines. Peak memory is the maximum runtime memory footprint in MB; higher throughput and lower memory are preferred.

Table 8 and Figure 3 report the 10k-scale points under explanation-aware accounting. The relevant comparison here is not raw forward speed alone, but end-to-end inference-plus-explanation throughput once post-hoc explanation cost is charged to the black-box baselines. Under that accounting, AnchorMoE reaches **122.8** iter/s for the 10k feature-dimension point (vs. **30.4** for DLinear), and **57.1** iter/s for the 10k sequence-length point (vs. **6.7** for TSLANet). These results therefore support an operational efficiency advantage in

Table 1: Real-data faithfulness evaluation. The top block reports overall means and mean ranks; the lower block details dataset-level metrics. Higher is better for I(20) and I(100); lower is better for rank. The best and second-best values in each row are highlighted.

Dataset / Stat.	Metric	AnchorMoE	AnchorMoE-gradient-free	IG	DeepLift	Grad-CAM	Random	Dynamask	FIT
Summary statistics									
Overall mean	I(20)	**0.6419**	**0.6338**	0.6189	0.6168	0.5891	0.5301	0.5390	0.5760
Overall rank	I(20)	**2.86**	3.43	**3.14**	3.86	4.71	6.29	6.00	5.71
Overall mean	I(100)	**0.8620**	**0.8667**	0.8384	0.8363	0.8278	0.8035	0.8284	0.8264
Overall rank	I(100)	**2.57**	**2.86**	3.71	4.43	5.86	6.29	4.57	5.71
Dataset-level values									
ArticularyWordRecognition	I(20)	0.6554	0.6625	**0.6808**	**0.6749**	0.5832	**0.6749**	0.5757	0.6361
ArticularyWordRecognition	I(100)	0.8772	0.8810	0.8812	**0.8963**	0.8599	**0.8938**	0.8812	0.8725
BasicMotions	I(20)	**0.5750**	**0.5938**	0.4625	0.5094	0.5521	0.4583	0.5083	0.4531
BasicMotions	I(100)	**0.8175**	**0.8675**	0.7354	0.7306	0.7792	0.7388	0.8092	0.7819
Cricket	I(20)	0.6597	0.6111	**0.6985**	**0.6997**	0.6140	0.6198	0.6024	0.6869
Cricket	I(100)	0.8993	0.8819	**0.9041**	**0.9078**	0.8703	0.8443	0.8760	0.8892
FordA	I(20)	0.6661	**0.6821**	0.6301	0.6412	0.6469	0.4620	**0.6920**	0.5631
FordA	I(100)	0.8678	**0.8766**	0.8401	0.8287	0.8541	0.7505	**0.8800**	0.8061
NATOPS	I(20)	0.6764	0.6708	**0.6792**	**0.6769**	0.6069	0.5315	0.4843	0.5486
NATOPS	I(100)	**0.8883**	**0.8808**	0.8744	0.8671	0.8517	0.8500	0.7591	0.8252
RacketSports	I(20)	**0.5576**	0.5206	0.4729	0.4482	**0.5414**	0.4238	0.4833	0.4512
RacketSports	I(100)	**0.7566**	**0.7541**	0.7054	0.7053	0.7194	0.6762	0.7278	0.6910
SpokenArabicDigits	I(20)	**0.7030**	0.6959	**0.7081**	0.6675	0.5790	0.5405	0.4271	0.6928
SpokenArabicDigits	I(100)	**0.9274**	0.9251	**0.9283**	0.9180	0.8600	0.8711	0.8658	0.9191

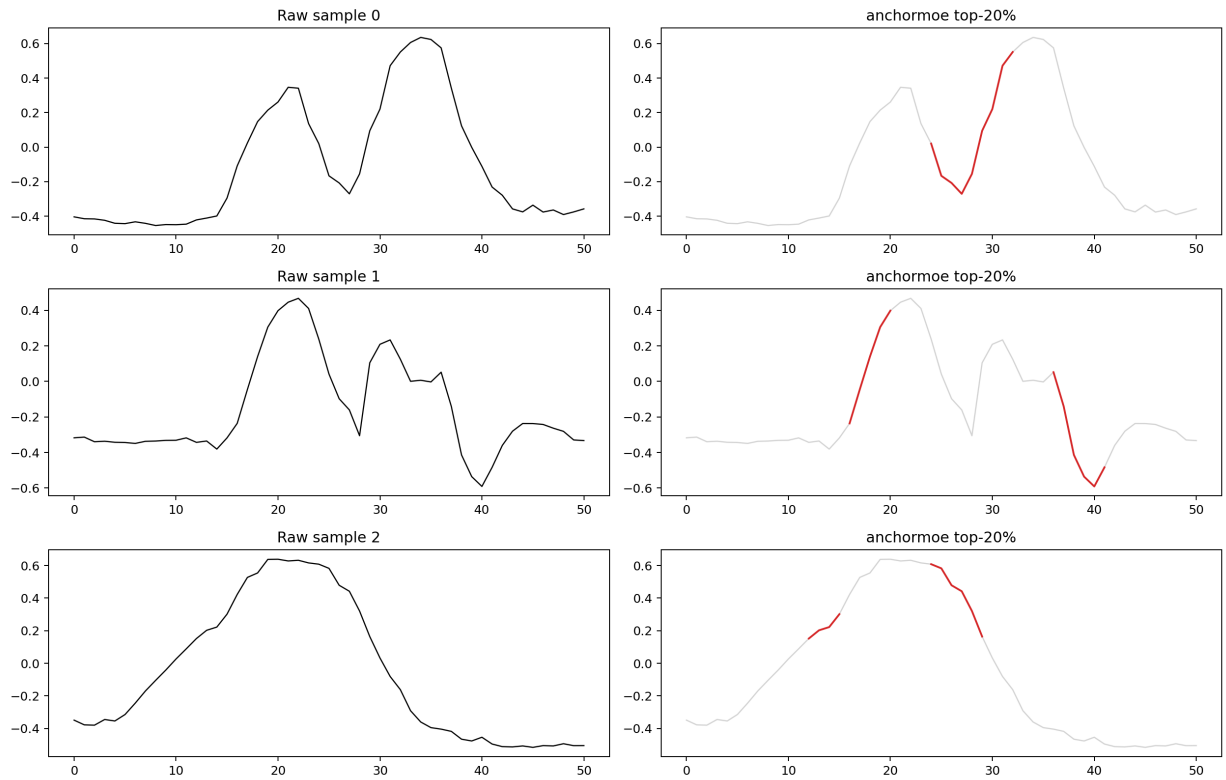


Figure 1: Representative NATOPS case study. Temporal evidence highlighted by AnchorMoE is compact and localized.

Table 3: Aggregate mechanism statistics across the 29-dataset diagnostic sweep.

Metric	Mean	Median	Min	Max
active_expert_rate	0.5388	0.5000	0.1667	1.0000
max_expert_share	0.7674	0.7578	0.4705	1.0000
effective_experts	1.8952	1.9053	1.0000	3.2933
κ_mean	0.1844	0.1662	0.0001	0.6128
κ_low_ratio	0.5653	0.6325	0.0000	1.0000
κ_high_ratio	0.0272	0.0002	0.0000	0.2603

Table 4: Per-dataset expert utilization. The model adapts to both distributed and concentrated routing regimes depending on the dataset.

Dataset	Active rate	Max share	Effective experts
Libras	1.0000	0.4932	3.2933
Cricket	0.6667	0.5594	3.0690
PhonemeSpectra	0.3750	0.4705	2.9461
BasicMotions	0.7500	0.5656	2.2905
Heartbeat	0.2500	1.0000	1.0003
MotorImagery	0.1667	1.0000	1.0000

Table 5: Per-dataset κ statistics. The gating mechanism adapts to varying sparsity regimes across different datasets.

Dataset	κ mean	Low ratio	High ratio
Libras	0.61276	0.07514	0.22591
Handwriting	0.44180	0.33335	0.26026
NATOPS	0.11251	0.81994	0.03575
SpokenArabicDigits	0.08893	0.86469	0.03999
Heartbeat	0.01114	0.97854	0.00000
MotorImagery	0.00007	0.99996	0.00000

Table 6: B2 routing diagnostics. Removing DPAR causes a severe 106.8% increase in inter-anchor redundancy.

Variant	Active rate	Max share	Effective experts	κ mean	Low ratio	High ratio	Redundancy
Full	0.4389	0.8525	1.4280	0.5319	0.1483	0.2606	0.3488
NoDPAR	0.5470	0.8283	1.4563	0.5600	0.1432	0.2865	0.7213
WeakDPAR	0.4278	0.8432	1.4518	0.5306	0.1528	0.2714	0.3788

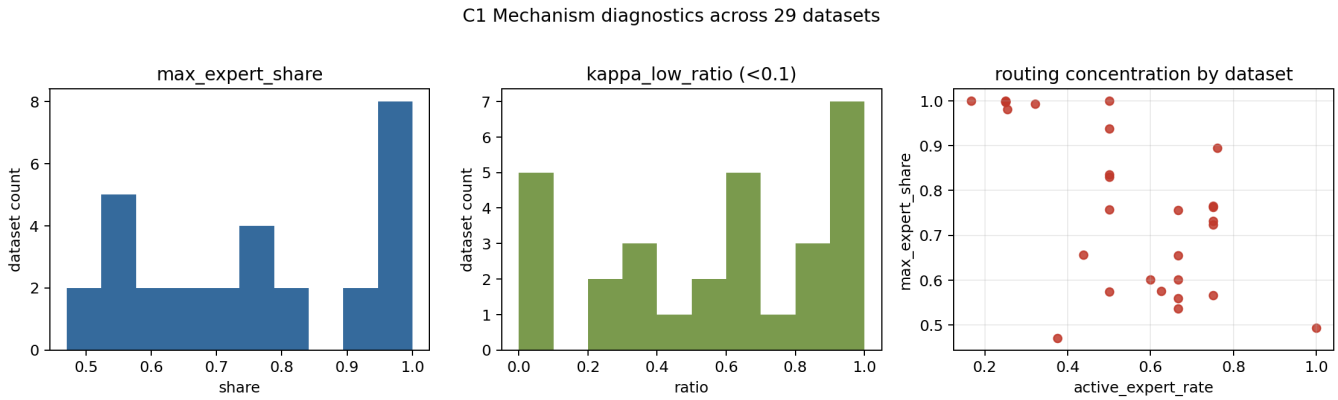


Figure 2: Mechanism diagnostics across 29 datasets, highlighting dataset-dependent routing concentration and non-trivial κ behavior.

the reported high-scale settings, rather than merely a favorable asymptotic argument.

Table 8: Explanation-aware throughput and peak memory for 10k-scale points. Throughput is measured in inference-plus-explanation iter/s (higher is better); memory in MB (lower is better).

Sweep	Point	AnchorMoE	Peak MB	Best baseline	Baseline	Rank
feature_dim	10000	122.822	535.1	DLinear	30.430	1
seq_len	10000	57.059	230.3	TSLANet	6.659	1

2. Interpretability on Real Data (W2)

Metric guide. For the real-data interpretability concern you raised, `I(20)` and `I(100)` again serve as the strict and full-budget temporal-faithfulness scores under the same protocol, so this section should be read as evidence about predictive temporal localization rather than full semantic interpretation.

We apply the same real-data faithfulness protocol discussed in Section 1. Please refer to Table 1 and Figure 1 for evidence supporting temporal faithfulness on real-world datasets.

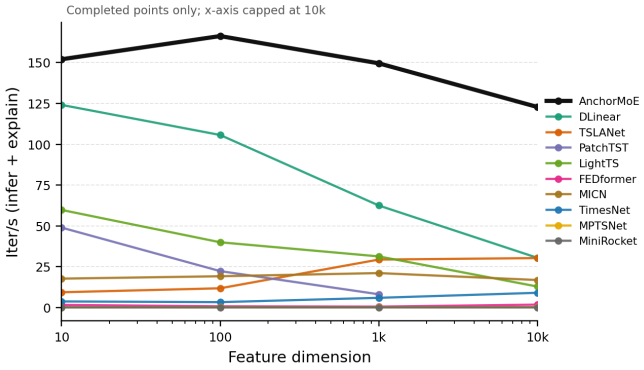
4 Additional Experiment Results to Reviewer ncgk

1. Tuning Complexity & Anchor Determination (W1, W3)

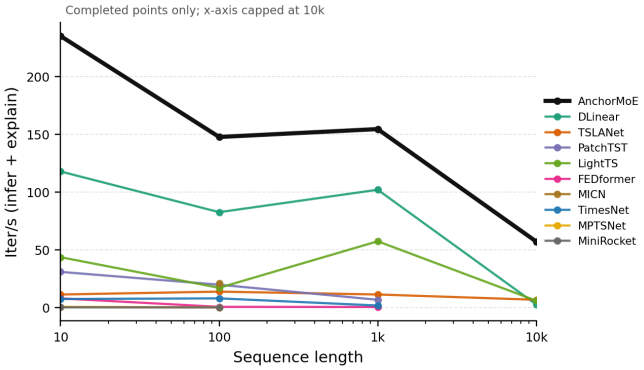
Metric guide. For the tuning factors you mentioned, anchor diversity, spectral weight, group count, and confidence regularization are the four scanned factors. In Table 10, `AOPCAcc` is an accuracy-under-perturbation score and `IoU` is overlap consistency; higher is better for both.

Table 7: Targeted internal-design ablations probing the proposed query, routing, and binding choices under a diagnostic setup. These block-wise results are not directly comparable to the 18-dataset overall averages reported in Table 2 of the main paper. Higher is better for all metrics.

Block	Variant	Mean Acc	Mean F1	Mean I(20)	Mean I(100)
B1	Full	0.9686	0.9685	0.9250	0.9348
B1	NoRelevance	0.7084	0.6687	0.8478	0.9348
B1	NoSpectral	0.9737	0.9736	0.9268	0.9348
B2	Full	0.9686	0.9685	0.9250	0.9348
B2	NoDPAR	0.9727	0.9726	0.9580	0.9348
B2	WeakDPAR	0.9734	0.9732	0.9518	0.9348
B3	Full	0.9509	0.9504	0.8953	0.9081
B3	AnchorMoE-gradient-free	0.9638	0.9638	0.9097	0.9051
B3	NoConfidence	0.9477	0.9475	0.8819	0.9078



(a) Feature-dimension sweep.



(b) Sequence-length sweep.

Figure 3: Explanation-aware throughput trends evaluated up to the 10k scale.

Table 9, Figure 4, and Table 10 address the practical tuning picture. The evidence here is not one of uniformly brittle behavior. In the reported sweeps, `anchor_div`, `spectral_weight`, and `num_groups` each admit a workable operating region, whereas `conf_lambda` is the clear exception and requires conservative tuning. On the more difficult NATOPS dataset, aggressive confidence regularization drives accuracy sharply from **0.9500** to **0.1667**. We therefore recommend a plateau-based heuristic for anchor-count selection: choose the smallest group count that already reaches the validation plateau. As shown in Table 10, the optimal `num_groups` consistently falls within a moderate range (**3**, **4**, or **5**).

Table 9: Sensitivity analysis summary. In the reported sweeps, `anchor_div`, `spectral_weight`, and `num_groups` show broader workable ranges, whereas `conf_lambda` is the most sensitive factor.

Parameter	Scanned values	Acc. range	Reading
anchor div.	0, 0.02, 0.05, 0.08, 0.1	0.9268–0.9332	Broadly workable in the current sweep.
spectral weight	0, 0.1, 0.3, 0.5, 0.7	0.9281–0.9374	Moderate values remain stable in aggregate.
num groups	2, 3, 4, 5, 6, 7, 8	0.9166–0.9722	Use the smallest count already on the plateau.
conf lambda	0, 0.005, 0.01, 0.02, 0.03	0.7976–0.9314	Most sensitive factor; NATOPS drops to 0.1667 under aggressive penalization.

2. Risk of Over-filtering & Safety-Critical Trade-offs (W2)

Metric guide. For the binding quantities you mentioned, `I(20)` and `I(100)` again indicate strict and full-budget temporal faithfulness, while κ mean / low / high summarize whether confidence weighting remains selective. In this table, a higher κ mean together with low-ratio collapse indicates that weak evidence is being admitted too uniformly.

Refer to Table 1 for the real-data faithfulness benchmark. Table 11 then isolates the binding question more directly: ablating confidence weighting (NoConfidence) forces an all-one gating regime and lowers mean `I(20)` from **0.8953** to **0.8819**. The implication is not that routing fails, but that routing alone is insufficient; a non-trivial confidence-weighted binding signal is still needed to prevent weak evidence from contributing on equal terms.

Table 10: Dataset-specific optimal `num_groups`. The optimum stays within a moderate range (3–5).

Dataset	Best num_groups	Acc	F1	AOPCAcc	IoU
ArticularyWord Recognition	3	0.9733	0.9733	0.7898	0.2820
Cricket	5	0.9722	0.9720	0.7000	0.2796
FordA	3	0.9356	0.9356	0.3461	0.2499
Heartbeat	4	0.7707	0.7674	0.4898	0.2769
NATOPS	4	0.9833	0.9833	0.6578	0.3750
SpokenArabic Digits	3	0.9900	0.9900	0.7773	0.3007

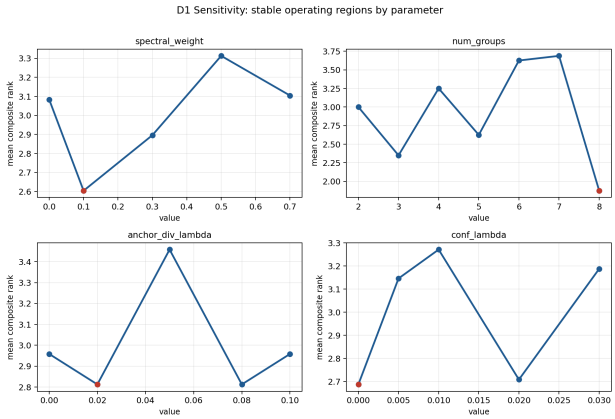


Figure 4: Stable operating regions for the primary structural hyper-parameters.

Table 11: Selective-binding diagnostics. NoConfidence yields an all-one gate and lower early-stage faithfulness.

Variant	I(20)	I(100)	κ	Low	High
Full	0.8953	0.9081	0.3833	0.0989	0.0504
AnchorMoE-gradient-free	0.9097	0.9051	0.2322	0.4188	0.0049
NoConfidence	0.8819	0.9078	1.0000	0.0000	1.0000